

Dibyendu Das

PhD Student in Mathematics

Indian Statistical Institute, Bangalore • Supervisor: Prof. Maneesh Thakur

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Algebraic Groups

Automorphism Groups

Associative Algebras

Research Interests

Algebraic Groups and related structures. My primary focus is on the algebraic and geometric properties of **automorphism groups of finite-dimensional associative algebras**, including their R -equivalence classes, solvability, and connections to isolated hypersurface singularities.

Publications & Preprints

[1] On R -Equivalence of Automorphism Groups of Associative Algebras (*Submitted*) [🔗](#)

[2] On Solvability of Automorphism Groups of Commutative Algebras (*Submitted*) [🔗](#)

Education

PhD in Mathematics

Indian Statistical Institute, Bangalore

Supervisor: Prof. Maneesh Thakur • Thesis area: Automorphism groups of algebras

2022 – Present

M.Math (Master of Mathematics)

Indian Statistical Institute, Bangalore

2019 – 2021

88.25%

B.Sc in Mathematics

St. Xavier's College, Kolkata

2016 – 2019

84.39%

Higher Secondary (Class XII)

Jangalpara BCKM High School (WBCHSE)

2014 – 2016

93.8%

Secondary (Class X)

KCS High School (WBBSE)

2014

89.5%

National Examinations & Entrance Results

Examination	Year	Rank / Result
IIT JAM Mathematics	2019	AIR 145
CSIR-NET Lectureship	2019	AIR 47
ISI M.Math Entrance	2019	AIR 6
GATE Mathematics	2021	AIR 339
CSIR-NET Mathematical Science (June)	2021	AIR 85
GATE Mathematics	2022	AIR 13
ISI JRF Entrance	2022	AIR 8
NBHM PhD Entrance	2022	Top 40

Also selected for: [INSPIRE Fellowship \(Stage 2\)](#) [JRF UGC NET 2021](#) [NBHM Doctoral Fellowship 2022](#)

Awards & Scholarships

INSPIRE Scholarship — DST, Govt. of India (Top 1% in Class XII)

2016 – 2020

ISI Scholarship — <i>Indian Statistical Institute</i>	2020 – 2021
Ram Ghosh Encouragement Award — <i>St. Xavier's College, Kolkata (70%+ in Semesters I–IV)</i>	2016 – 2018
Dhruva Jyoti Bhattacharya Memorial Gold Medal — <i>St. Xavier's College (Highest marks in Mathematics, B.Sc)</i>	2016 – 2019

Teaching Experience

Teaching Assistant — Algebraic Topology — <i>ISI Bangalore, B.Math 3rd Year (Instructor: Charanya Ravi)</i>	Fall 2023
Teaching Assistant — Probability III — <i>ISI Bangalore, B.Math 3rd Year (Instructor: Vivek Kumar)</i>	Fall 2024
Teaching Assistant — Topology I — <i>ISI Bangalore, B.Math 2nd Year (Instructor: Maneesh Thakur)</i>	Spring 2025
Teaching Assistant — Probability II — <i>ISI Bangalore, BSDS 2nd Year (Instructor: Antar Bandyopadhyay)</i>	Fall 2025
Teaching Assistant — Geometric Group Theory — <i>ISI Bangalore, B.Math 3rd Year (Instructor: Maneesh Thakur)</i>	Spring 2026
Teaching Assistant -Algebraic Number Theory — <i>ISI Bangalore, B.Math 3rd Year, M.Math 2nd Year (Instructor-Maneesh Thakur)</i>	Fall 2026
Evaluator — Regional Mathematical Olympiad (Karnataka) — <i>Homi Bhabha Centre for Science Education</i>	2023

Lecture Series (Advanced Graduate)

Affine Algebraic Groups and their Structures — <i>ISI Bangalore</i>	Jan – Jun 2023
Triangular Algebras and Book of Involution — <i>ISI Bangalore</i>	Jan – Aug 2024
Schemes — <i>ISI Bangalore</i>	Aug – Nov 2025
Theory of higher degree forms and their automorphism groups — <i>ISI Bangalore</i>	Jan 2026 – May 2026

Research Talks

Algebraic Groups and Automorphism Groups of Algebras — <i>PDF-RS Symposium, ISI Bangalore</i>	Oct 2023
On Automorphism Groups of Algebras — <i>JRF to SRF Evaluation, ISI Bangalore</i>	May 2024
R-Equivalence on Automorphism Groups of Associative Algebras — <i>RDGT 2025, IIT Bhubaneswar</i>	Jul 2025
Solvability of Automorphism Groups and Isolated Hypersurface Singularities — <i>PDF-RS Symposium, ISI Bangalore</i>	Mar 2026
On R-Equivalence of Automorphism groups of Associative Algebras — <i>Colloquium Talk — ISI Bangalore</i>	Mar 2026
Solvability of Automorphism groups of Commutative Algebras — <i>PGTC 2026, University of Manchester, UK.</i>	June 2026
Solvability of Automorphism groups of Commutative Algebras and Isolated Hypersurface Singularities — <i>GGI Seminar 2026, University of Essex, UK.</i>	July 2026

Conferences & Schools Attended

- AIS on Birational Geometry, Chennai Mathematical Institute (NCM) May–Jun 2023
- Zariski Dense Subgroups, Number Theory and Applications, ICTS Bangalore Jan 2024
- AIS on Representation Theory of Quivers and BGG Category $\mathcal{O}$, IIT Kanpur (NCM Online) Dec 2024
- Recent Developments in Group Theory (RDGT 2025), IIT Bhubaneswar Jul 2025
- Post Graduate Group Theory Conference (PGTC 2026), University of Manchester June–July 2026
- ATM School on Advanced Commutative Algebra, IIT Gandhinagar (NCM) Jan 2021
- Zoom Algebraic Geometry Seminar, University of Essex, UK Jun 2021
- Annual Seminar “ANALYTICA”, St. Xavier’s College, Kolkata 2017, 2018, 2019

Undergraduate Project

A Study on Solvable Groups — B.Sc. Dissertation, St. Xavier’s College (Supervisor: Prof. Rabiul Islam) Dec 2018 – Jun 2019

Studied solvable and nilpotent groups and applied representation theory to establish Burnside’s theorem.